Viscosity measurements of samples from the volcanic chain off Mayotte

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Motivation

In June 2018 an unexpected eruption occurred 50 km off Mayotte. It was only one year later that sampling of this area was carried out, largely due to its submarine settings. Our study focuses on five samples from Fani Maoré (two basanites, one tephriphonolite, and two phonolites). It examines magma viscosity and provides insights into volcanic parameters and the associated eruption hazards. Viscosity helps to identify the type of eruption involved, as well as magma ascent mechanisms, such as flow velocity. Given the proximity to Mayotte, it is important to study these zones to assess potential hazards, especially since degassing is still active today.

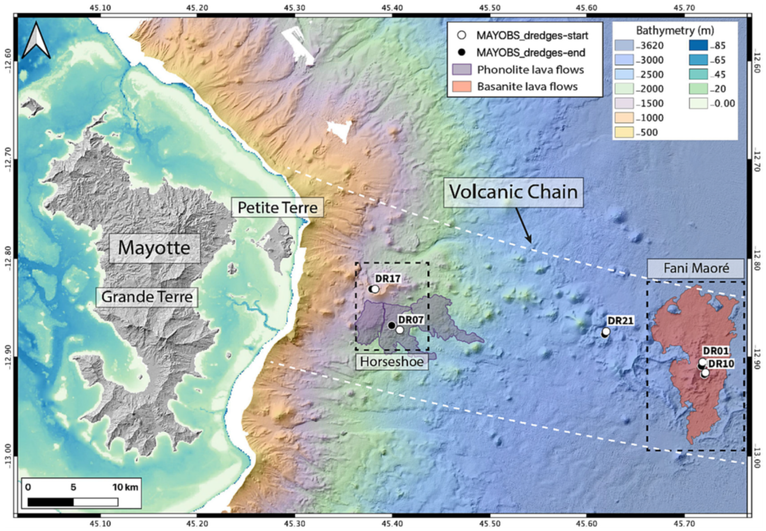

Figure 1. Geological map of the submarine volcanic chain off mayotte showing the location of the samples.

This map (figure 1) provides an overview of the distribution of the sampling points along the volcanic chain, obtained through dredging. It also shows the bathymetry of the area, with samples located at depths between 2,000 and 3,000 meters. Analyses indicate a source near DR17 and DR07, at the level of the Horseshoe volcanic structure.

Experimental methods

The study has been conducted with five samples of volcanic deposits collected along the seafloor, between Mayotte and Fani Maoré via the Horseshoe. They are basanites, phonolites and tephriphonolites. To study their major elements composition, ICP-AES measurements have been conducted. The sample's degree of polymerization (Equation 1) was calculated as well.

$$\frac{NBO}{T} = \frac{(2O) + (4T)}{T} \tag{1}$$

The viscosity has alsoo been calculated at high and low temperatures. A viscometer has been used for the high temperature measurements while the low temperaure viscosity has been assessed with a creep apparatus. Finally, nanoliths have been detected using Raman spectroscoy methods.

Results

measurements

The TAS diagram (Fig. 2) orders the rocks according to their SIO2 and NaO2 + K2O content. The samples from Mayotte are grouped in the Tephrite, Tephriphonolite and Phonolite facies; they follow an alkaline evolutionary trend which corresponds to a phase of magma differentiation. The more the samples are classified in the Basanite facies, the less differentiated they are (the lower the content of SiO2, NaO2 and K2O).

Results

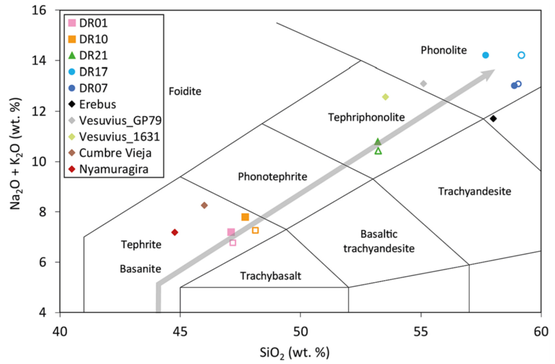

Figure 2. TAS diagram

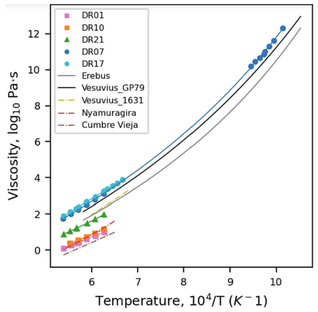

Figure 3. Sample viscosity as a function of inverse temperature

The relationship between viscosity and temperature is almost linear : at super liquidus temperature, results show that basanites have the lowest viscosity (maximum of 1.16 $\log_{10} \cdot Pa$ at 1596 K and 0.34 $\log_{10} \cdot Pa$ at 1810 K) while phonoliths have the highest viscosity(minimum of 3.10 $\log_{10} \cdot Pa$ at 1597 K and 1.75 $\log_{10} \cdot Pa$ at 1860 K). The difference is two orders of magnitude. Tephriphonolite melts have an intermediate 1.97 $\log_{10} \cdot Pa$ viscosity at 1596 K and 0.89 $\log_{10} \cdot Pa$ at 1858 K. It was only possible to measure low temperature viscosity for one sample : DR07. Its viscosity varies from 12.30 $\log_{10} \cdot Pa$ at 968 K to 10.19 $\log_{10} \cdot Pa$ at 1058 K.

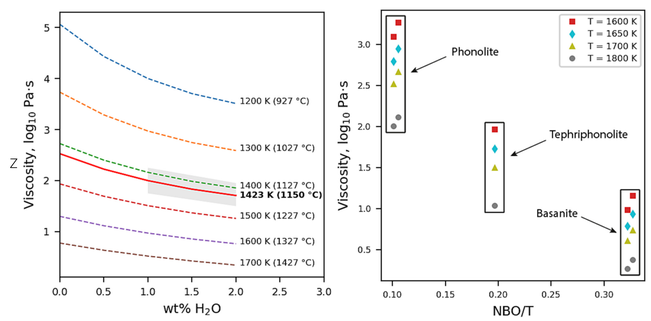

Figure 4. (A) Isothermal viscosity as a function of water concentration for the DR10 basanite (B) Viscosity as a function of K/(NBO/T) ratio

Considering the water content, the viscosity is lower when adding water, and this difference is bigger at low temperatures. This is shown in figure 4A. There is a linear relationship between the measured viscosity and the calculated polymerization degree at a given temperature, as shown in figure 4B. The basanite samples present the highest degree of polymerization (around 0.35) while having the lowest viscosity, maximum 1.25 (Fig 4B) while Phonolite samples show the highest viscosity (minimum of 2.0 $\log_{10} \cdot Pa$) and the lowest polymerization (around 0.10). In between them, Tephriphonolites have an intermediate degree of polymerization and viscosity.

Results

Models

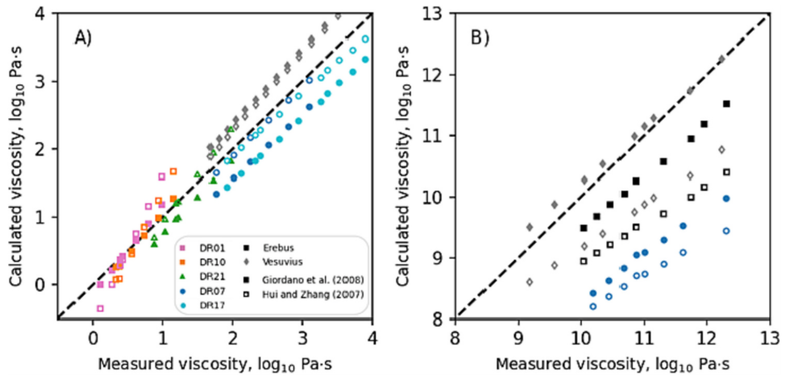

Figure 5. Calculated viscosity (from the paper [1] and [2] models) depending on the measured viscosity, for super-liquidus temperatures (A) and undercooling (B).

Figure 5 shows the values of the calculated viscosity depending on the measured viscosity. If the models fit the reality, the data should lign up along a straight line of equation $y = x$. For super-liquidus and high viscosities, we see that the model corresponds to the measurements, but not as much as for lower viscosities. For the undercooled temperatures, the calculated viscosity tend to underestimate the values, relatively to the measurements There are two reasons explaining these observations: For the undercooled temperatures, there are "nonlinear variations in melt configurational entropy" [3] resulting into variation in melt composition, which can impact the viscosity. Also, the [1]'s model was calibrated for alkaline basanite to phonolite data, which can impact the calculated viscosity for such rocks.

Nanoliths

The presence of nanoliths in magma can involve a modification of natural viscosity. Indeed, nanoliths are micro crystals (in the order of one nanometer: in the article the size is ranging from 5 to 30 nm), which come from magma or volcanic glass. The formation of nanoliths appears as soon as cooling occurs, they come from rapid crystallization. They can, as mentioned previously, increase the viscosity of magma, but also increase outgassing and trigger explosive behaviors. We still try to understand if nanoliths have had an impact during the Fani Maoré eruption.

Conclusions

- Existing models are well suited to high temperatures, but tend to underestimate the values for lower temperatures.
- The study reveals the importance of the viscosity in the eruption mecanisms and risks.
- Those measurements highlight the need for better eruptive models for the alkaline magmas in general.
- Nanoliths increase the viscosity and can trigger explosive eruptions behaviors, raising questions that deserve further investigation.

References

[1] Daniele Giordano, James K. Russell, and Donald B. Dingwell. Viscosity of magmatic liquids: A model. 271(1):123–134.

[2] Hejiu Hui and Youxue Zhang. Toward a general viscosity equation for natural anhydrous and hydrous silicate melts. 71(2):403–416.

[3] Pauline Verdurme, Charles Le Losq, Oryaëlle Chevrel, Salomé Pannefieu, Etienne Médard, Carole Berthod, Jean-Christophe Komorowski, Patrick Bachèlery, Daniel R. Neuville, and Lucia Gurioli. Viscosity of crystal-free silicate melts from the active submarine volcanic chain of mayotte. 620:121326.

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